Sequence representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable sequence and real-position record.

Trace order,apply masks in calculations and explain unknown/truncated information.

Keep one habit: W1–W3/W5 → selected new sequence schema → project encoding ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use fictional robot event logs whose symbols are GO, TURN and STOP. The task asks whether the final real event is STOP. This is a representation exercise, not a claim that an event log proves a robot is safe.

  GO then STOP and STOP then GO contain the same symbols but have different final events.

- A count vector discards order, so no deterministic rule using only the first two count vectors can distinguish their answers. A **sequence** of IDs preserves order if we retain each position.

  Declare the vocabulary: PAD = 0, UNK = 1, GO = 2, TURN = 3 and STOP = 4. PAD is reserved for added storage positions; UNK represents a real but unrecognised symbol.

- IDs are names, not measured amounts. STOP having ID 4 does not make it twice as much as GO with ID 2. Changing the vocabulary requires changing every consumer consistently.

  Do not interpret numerical distance between IDs as semantic similarity.

Augmentation and leakage

> **My remembered idea, example and correction**

Pooling and spatial information

> **My remembered idea, example and correction**

Learned representations

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Log	Counts in GO TURN STOP order	Final event	Task answer
GO STOP	[1, 0, 1]	STOP	Yes
STOP GO	[1, 0, 1]	GO	No
GO TURN STOP	[1, 1, 1]	STOP	Yes

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main vocabulary:PAD0, UNK1, GO2, TURN3, STOP4. Store four positions with right padding. GO STOP encodes [2, 4, 0, 0], mask[1, 1, 0, 0]; STOP GO gives [4, 2, 0, 0]and the same mask. TURN STOP gives [3, 4, 0, 0], real length 2, last real index 1 with STOP4. GO TURN STOP and STOP TURN GO have equal counts[1, 1, 1]but different final events.

  WAIT GO and PAUSE GO both encode [1, 2, 0, 0], losing the unknown symbol identity while keeping a real position. Empty/overlong logs and literal reserved PAD/UNKstrings are rejected; actual strings are assumed. Five-event GO TURN GO TURN STOPends in STOP, but its first-four prefix ends in TURN. Sensor values [2, 4, 0, 0]with mask[1, 1, 0, 0]have mean 3, not 1.5.

  Values[0, 4, 0, 0]with that mask have mean 2, not zero-filter mean 4or all-slot mean 1. Token IDs are names, not quantities to average. Separate picture LEFT2/RIGHT3 uses its own vocabulary. The independent task supplies a new vocabulary, length and measurements.

- Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main vocabulary:PAD0, UNK1, GO2, TURN3, STOP4. Store four positions with right padding. GO STOP encodes [2, 4, 0, 0], mask[1, 1, 0, 0]; STOP GO gives [4, 2, 0, 0]and the same mask. TURN STOP gives [3, 4, 0, 0], real length 2, last real index 1 with STOP4. GO TURN STOP and STOP TURN GO have equal counts[1, 1, 1]but different final events.

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  Values[0, 4, 0, 0]with that mask have mean 2, not zero-filter mean 4or all-slot mean 1. Token IDs are names, not quantities to average. Separate picture LEFT2/RIGHT3 uses its own vocabulary. The independent task supplies a new vocabulary, length and measurements.

- Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main vocabulary:PAD0, UNK1, GO2, TURN3, STOP4. Store four positions with right padding. GO STOP encodes [2, 4, 0, 0], mask[1, 1, 0, 0]; STOP GO gives [4, 2, 0, 0]and the same mask. TURN STOP gives [3, 4, 0, 0], real length 2, last real index 1 with STOP4. GO TURN STOP and STOP TURN GO have equal counts[1, 1, 1]but different final events.

  WAIT GO and PAUSE GO both encode [1, 2, 0, 0], losing the unknown symbol identity while keeping a real position. Empty/overlong logs and literal reserved PAD/UNKstrings are rejected; actual strings are assumed. Five-event GO TURN GO TURN STOPends in STOP, but its first-four prefix ends in TURN. Sensor values [2, 4, 0, 0]with mask[1, 1, 0, 0]have mean 3, not 1.5.

  Values[0, 4, 0, 0]with that mask have mean 2, not zero-filter mean 4or all-slot mean 1. Token IDs are names, not quantities to average. Separate picture LEFT2/RIGHT3 uses its own vocabulary. The independent task supplies a new vocabulary, length and measurements.

- Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main vocabulary:PAD0, UNK1, GO2, TURN3, STOP4. Store four positions with right padding. GO STOP encodes [2, 4, 0, 0], mask[1, 1, 0, 0]; STOP GO gives [4, 2, 0, 0]and the same mask. TURN STOP gives [3, 4, 0, 0], real length 2, last real index 1 with STOP4. GO TURN STOP and STOP TURN GO have equal counts[1, 1, 1]but different final events.

  WAIT GO and PAUSE GO both encode [1, 2, 0, 0], losing the unknown symbol identity while keeping a real position. Empty/overlong logs and literal reserved PAD/UNKstrings are rejected; actual strings are assumed. Five-event GO TURN GO TURN STOPends in STOP, but its first-four prefix ends in TURN. Sensor values [2, 4, 0, 0]with mask[1, 1, 0, 0]have mean 3, not 1.5.

  Values[0, 4, 0, 0]with that mask have mean 2, not zero-filter mean 4or all-slot mean 1. Token IDs are names, not quantities to average. Separate picture LEFT2/RIGHT3 uses its own vocabulary. The independent task supplies a new vocabulary, length and measurements.

- The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

  Calculate their **mask**ed mean and the incorrect mean over all five slots.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a fresh schema independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **sequence** representations **evidence** sheet to Image-pattern Network Investigation. Show how you can encode an ordered **sequence** with an explicit vocabulary.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Expose an order collision for **Sequence** representations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Sequence** representations. Proposed explanation: "Larger **token ID**s mean more of a concept". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

  Calculate their **mask**ed mean and the incorrect mean over all five slots.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Distinguish **padding** positions from meaningful zero sensor readings.

- Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/25

Sequence representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Encode real tokens

LEFT and RIGHT occupy positions 0 and 1.

↓

2. Mark padding

Positions 2 and 3 are storage padding; mask marks them 0.

↓

3. Compare sensor example

Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

CHANGE ONE THING

A missing sequence length causes every numeric 0 to be treated as padding in the sensor example.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Distinguish **padding** positions from meaningful zero sensor readings.
- Which labelled step matches this detail? Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.
- 1. Encode real tokens
- 2. Compare sensor example
- 3. Mark **padding**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare sensor example Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Changed-case question and optional hint

- Changed-case practice: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The first real sensor reading is incorrectly discarded.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Repair and retain the record

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

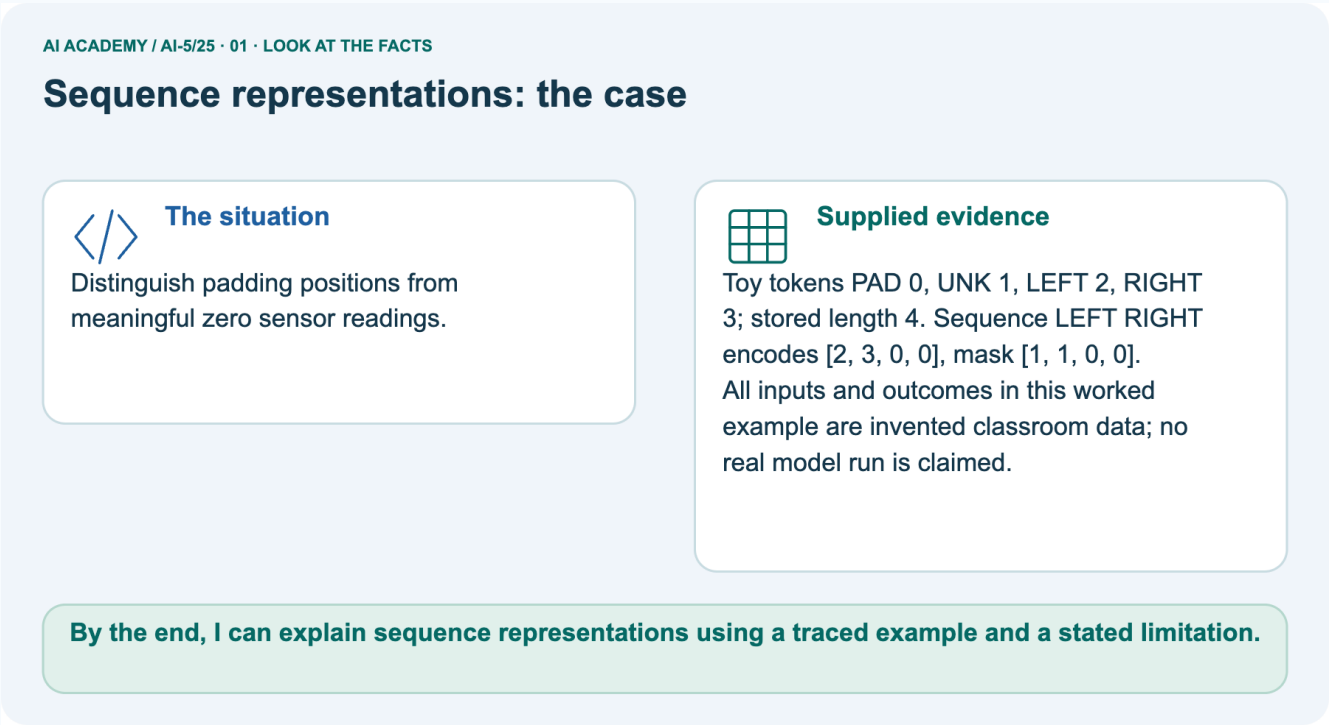
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Sequence representations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

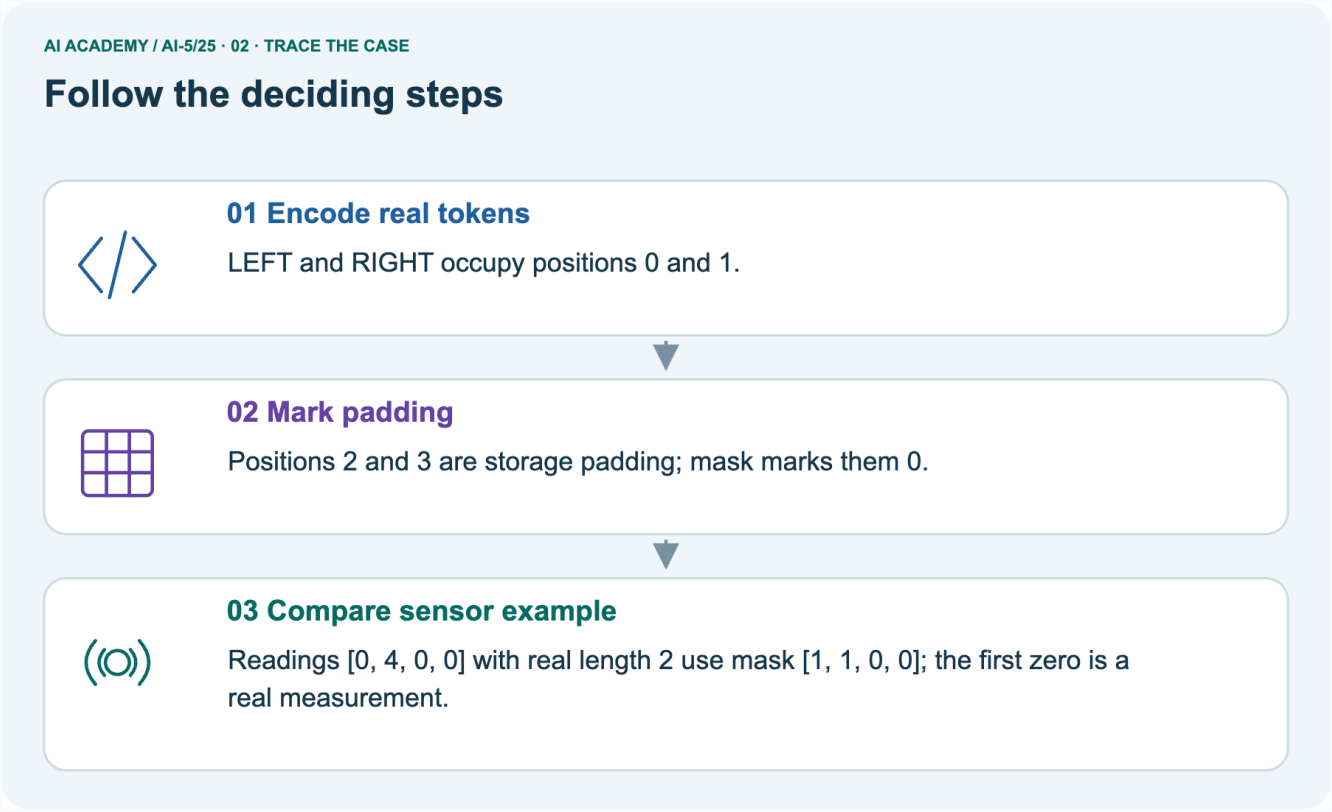
Distinguish **padding** positions from meaningful zero sensor readings. Supplied facts: Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Encode real tokens: LEFT and RIGHT occupy positions 0 and 1. → Mark **padding**: Positions 2 and 3 are storage **padding**; **mask** marks them 0. → Compare sensor example: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A padded log still has two real events

AI ACADEMY / AI-5/25 · 03 · INSPECT THE LESSON MODEL

A padded log still has two real events

Stored position	Token ID	Mask
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

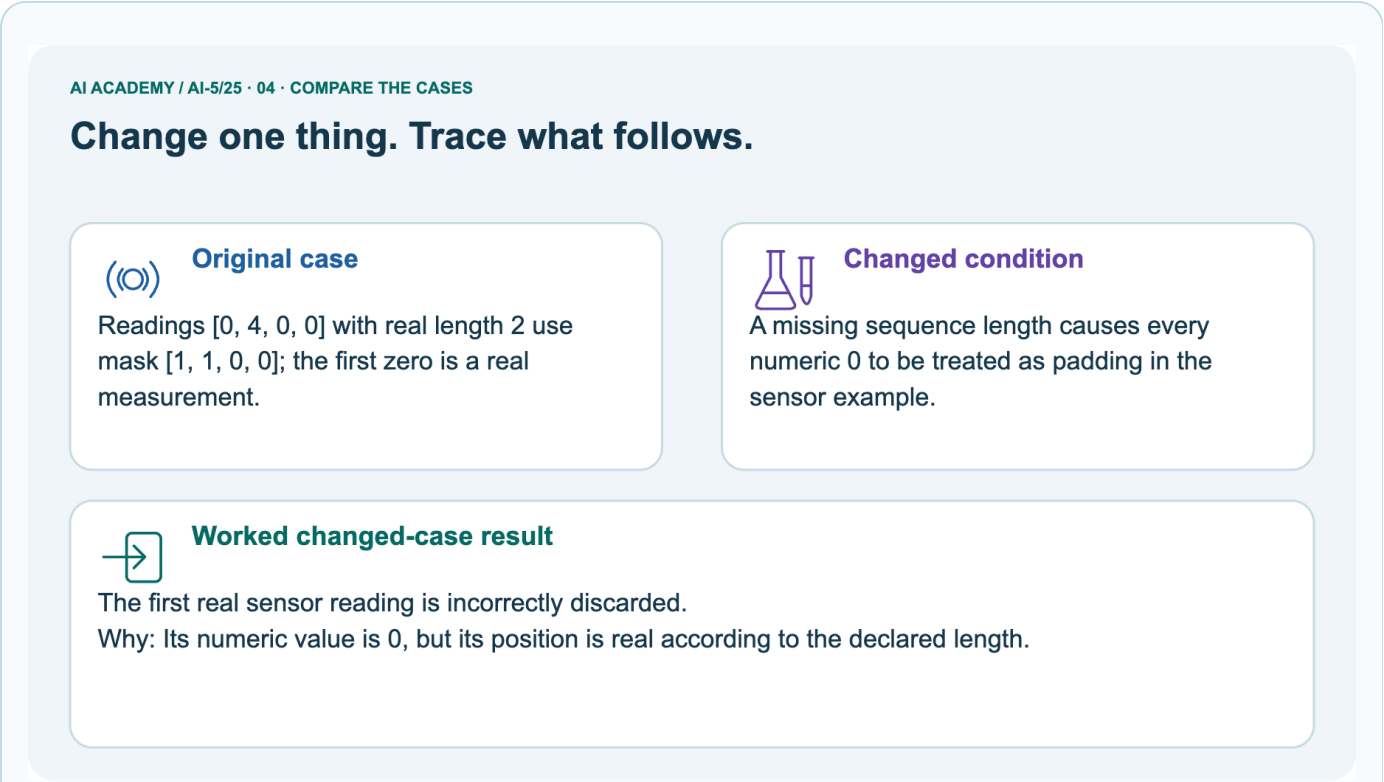
Follow the labelled picture with the matching explanation. Open the vector diagram

A padded log still has two real events. 0 | 2 meaning GO | 1 real. 1 | 4 meaning STOP | 1 real. 2 | 0 meaning PAD | 0 placeholder. 3 | 0 meaning PAD | 0 placeholder

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

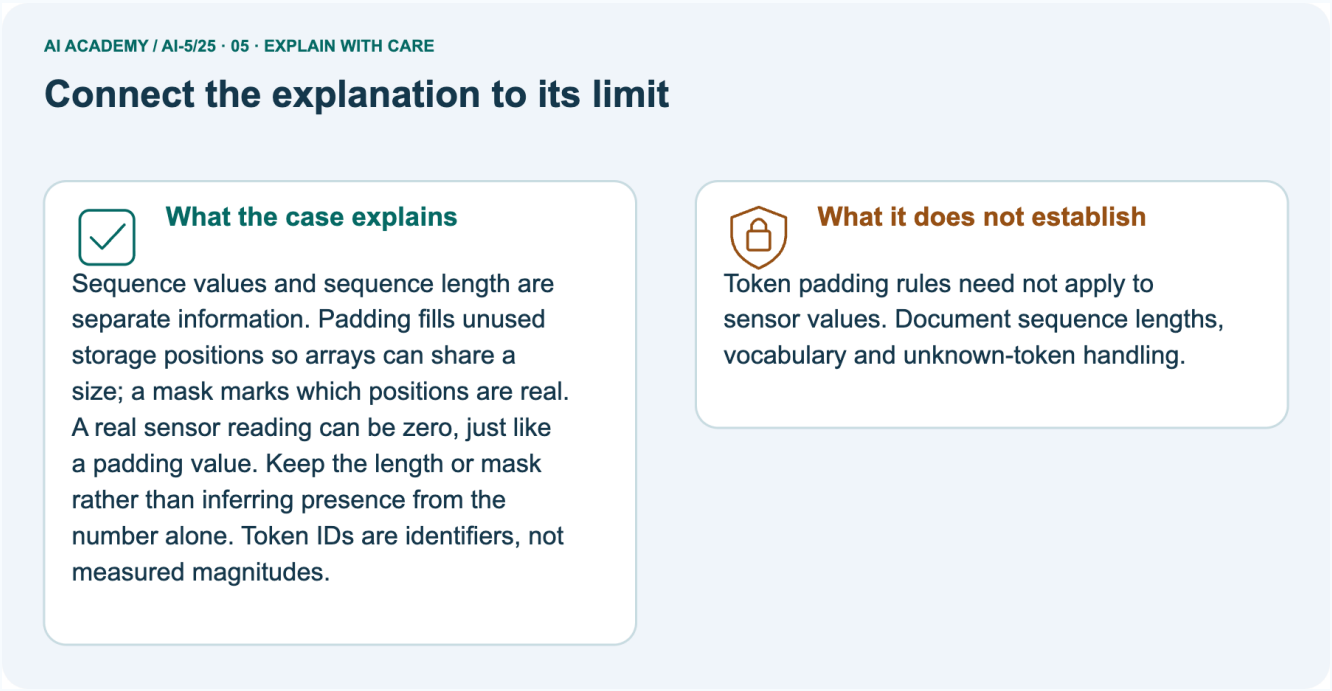
Changed condition: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Expected result: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

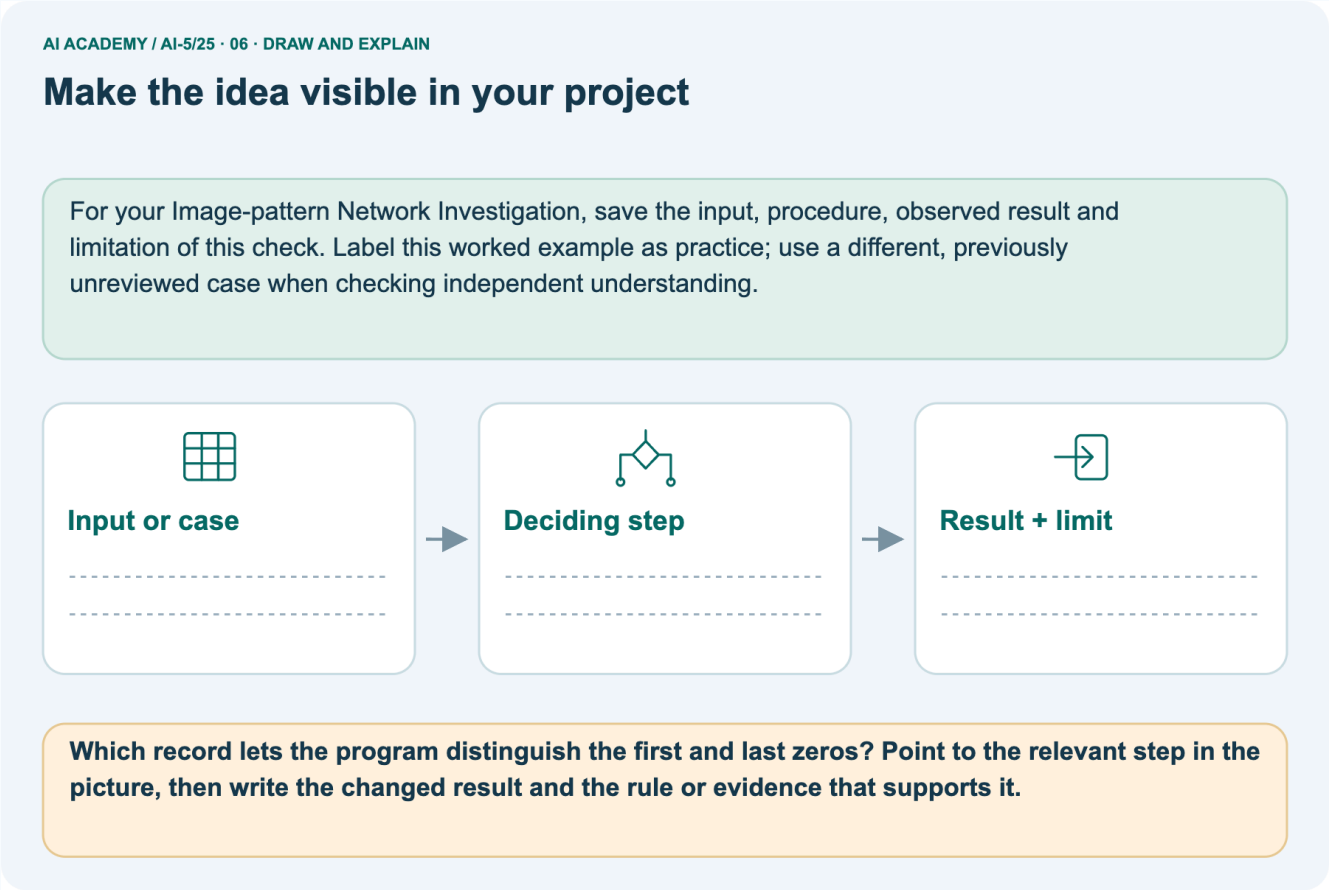
A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Limit: Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sequence representations: the case

02. Follow the deciding steps

03. A padded log still has two real events

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.